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EffiSwinNet: A Cross-Architecture Deepfake Detection with Style GAN Generated Faces

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***Abstract* — The threat of deepfakes to our security has been present for a while. The development of Generative Adversarial Networks (GANs), a network that assists in creating new datasets from existing datasets based on specific traits and statistics, has made these digitally produced images unrecognizable to human sight. Malicious activities including identity theft and cyberbullying have been carried out using these deepfakes, which contributes to cybercrime. The goal of this study is to use Shifting Window Transformer to identify deepfake images produced by GANs. Contrary to vision transformers (ViT), this transformer uses a shifting window-based Transformer block that helps to integrate picture patches and simplify the transformer model. It can also do feature level extraction similar to CNN. Three methods were employed in this study to detect deepfake photos using Swin-Transformer: Base Swin-Transformer, Swin-Transformer utilizing Transfer Learning, and Hybrid-Swin Transformer. StyleGAN-based deepfake photos from a dataset of 140K genuine and fake faces were employed. With an accuracy of 99% and a cross-entropy loss of 0.25, the hybrid model of EfficientNetB0+SwinTransformer outperformed the other two tests.**

Keywords—Deepfake detection, Vsion Transformers, EfficientNet, Swin-Transformer, StyleGAN

I. INTRODUCTION

The rapid advancement of digital media technologies has led to the proliferation of deepfake content, raising significant concerns regarding authenticity and trustworthiness. Recent studies have highlighted the challenges associated with distinguishing between computer-generated imagery (CGI) and authentic images, particularly in the context of deepfake technology. Deep fakes are a phrase derived from the technology that employs deep learning to generate fabricated but highly realistic media content like photos and videos. With the advancement in generative adversarial networks (GANs) generating these images has become the most common methods [1] .These hyper-realistic synthetic images have often been misused to spread misinformation [2] , manipulate public perception [3] , or impersonate individuals in harmful ways . A particularly troubling trend is the use of deepfake technology to target individuals especially public figures or private citizens in ways that violate their privacy, dignity, and security [4] . Banks, cryptocurrency exchanges, and other organizations that increasingly rely on picture and video identification to adhere to KYC laws face a danger [5] . Fake identities are also used for sentiment manipulation in financial market that can lead to various fraud and extortion [6] .

Deepfakes are high-fidelity synthetic data that are basically based on GAN, which consist of two networks working in conjunction with one another [7] [8] . These two network are Generator and Discriminator that compete with each other. For a given collection of genuine photographs, the generator's goal is to create similar real images by feeding it noise; the discriminator's job is to identify the generated image as real. To increase its capabilities, the model makes use of back propagation. GANs are especially good at creating synthetic faces that are visually identical to real ones because of this adversarial training process, which continues until the Generator generates data that is indistinguishable from authentic content and the Discriminator is unable to consistently tell the difference. Many recent research approaches has oversimplified the deep fake detection by treating the problem as a binary classification task. Style GAN is a Face Synthesis technique which is widely popular to create non-existent real faces [9] . Developed by NVIDIA, StyleGAN introduced a new generator design that enables control over the visual attributes of generated images through intermediate latent space manipulation and its architecture progressively grows the image resolution during training, allowing it to produce extremely detailed and high-quality facial images , with this approach in our paper we have also used StyleGAN to create synthetic images.

For a variety of computer vision tasks, Vision Transformers (ViTs) have become a potent substitute for conventional convolutional neural networks (CNNs) in recent years [10]. ViTs use a fully attention-based method[11], processing images as patches and directly learning global dependencies through self-attention, in contrast to CNNs, which rely on local receptive fields and spatial hierarchies[12]. The original Vision Transformer architecture showed that transformer-based models can perform better than CNNs [13] on tasks like segmentation, object detection, and image classification when pre-trained on sufficiently large datasets. This paradigm change has sparked a lot of interest in creating ViT variations that are more effective and scalable. Among these variations, the Swin Transformer enhances scalability and computational efficiency by introducing window-based multi-head self-attention and hierarchical feature encoding[14]. Swin Transformers are able to describe long-range dependencies while achieving linear computational complexity[15] in relation to image size by limiting self-attention computation to non-overlapping local windows and using a shifted windowing approach across layers. Additionally, it has been shown that combining the CNN based pretrained models like ResNeT 18, VGG 16, AlexNet [16] strengthens the potential of deepfake detection. Another work by [17] proposes a MobileNetV3 based model fine tunning with an integration of self attention layer which directly improves the model capacity in identifying the artifacts manipulated during the deepfake generation. Similarly, [18] proposes an another variation of CNN based model, in their work a fine-tuned Efficient Net model with integration of self-attention layer has been used to identify complex long dependencies and finding the in between relation with multiple unconnected regions of an image. Building on this, our work introduces a hybrid deep learning architecture that combines the representational strength of EfficientNetB0 with the global attention modeling capabilities of the Swin Transformer. While ViTs like Swin Transformer provide strong contextual understanding through self-attention mechanisms, EfficientNetB0 contributes efficient feature extraction with fewer parameters and lower computational cost. By integrating these two models, we aim to enhance deepfake detection performance, particularly for high-fidelity synthetic facial images generated by StyleGAN. This hybrid approach leverages both convolutional and transformer-based representations to improve accuracy, scalability, and robustness in real-world detection scenarios. Hence, the main contribution of this paper are as follows:

1. We propose a novel multi-output EffiSwinNet B0 architecture for efficient local feature extraction and integrating it with a Swin Transformer for capturing global contextual dependencies, optimizing both accuracy and computational cost.
2. The model is tailored to detect high-fidelity synthetic faces generated by StyleGAN, leveraging both local and global features to identify subtle artifacts in fake images.
3. EffiSwinNet outperforms baseline models in accuracy and loss, and demonstrates strong generalization on unseen datasets, confirming its robustness in real-world deepfake detection.

The remainder of the document is organized as follows: A significant and pertinent overview of the literature is included in Section 2, followed by Data and Methodology in Section 3, Results and Discussion in Section 4, and a conclusion to the study in Section 5.

II. LITERATURE REVIEW

Recent studies have focused on developing effective methods to detect deepfakes, especially those generated using GANs. Researchers have explored various models, including CNNs, Vision Transformers, and hybrid approaches, to improve accuracy, efficiency, and performance on real-world datasets. Nawaz et al.[19] propose a deep learning-based method called AUFF-Net , a unified framework for detecting FaceSwap (FS) and Face-Reenactment (FR) deepfakes. They employed Inception-Swish-ResNet-v2 model for spatial feature extraction and a Bi-LSTM network for capturing temporal dependencies. Experiments conducted on the FaceForensics++ dataset yielded average accuracies of 99.21% for FS and 98.32% for FR detection. Shahin et al [20] harness the potential of Vision Transformer (ViT) in conjunction with Convolutional Autoencoders for image analysis and deepfake detection. The proposed models yield excellent accuracy rate of approximately 87%. Usmani et al [21] propose a shallow vision transformer for deepfake detection which uses an attention mechanism with a multi-head attention module. The proposed model is shallow as it has 16.48 times fewer parameters and approx 2.97 times fewer FLOPS than the baseline vision transformer. Experiments on the Real Fake Face (RFF) and Real and Fake Face Detection (RFFD) datasets show that the model can achieve an accuracy of 92.15% and 88.52% respectively,. Soumya et al. [22] examines the SWIN Transformer which uses various self-attention mechanisms and advanced features, it relies on the concept of shifted windows during the processing of the images and is considered more effective than the traditional CNN methods. The results show that the SWIN Transformer-based method performs better in recognizing deep fake images. The accuracy is found to be 97.91% for CelebDF dataset and 95.71% for FF++ dataset. The Log Loss was calculated to be 0.034 for CelebDF dataset and 0.1573 for FF++ dataset.

Although Vision Transformer (ViT)-based models have gained attention for their ability to capture global context through self-attention, CNN-based models remain widely used due to their computational efficiency and strong performance on smaller datasets. Several studies continue to explore CNN architectures such as VGG, ResNet, and EfficientNet for deepfake detection, particularly in scenarios where transformer models may be resource-intensive. Ashani

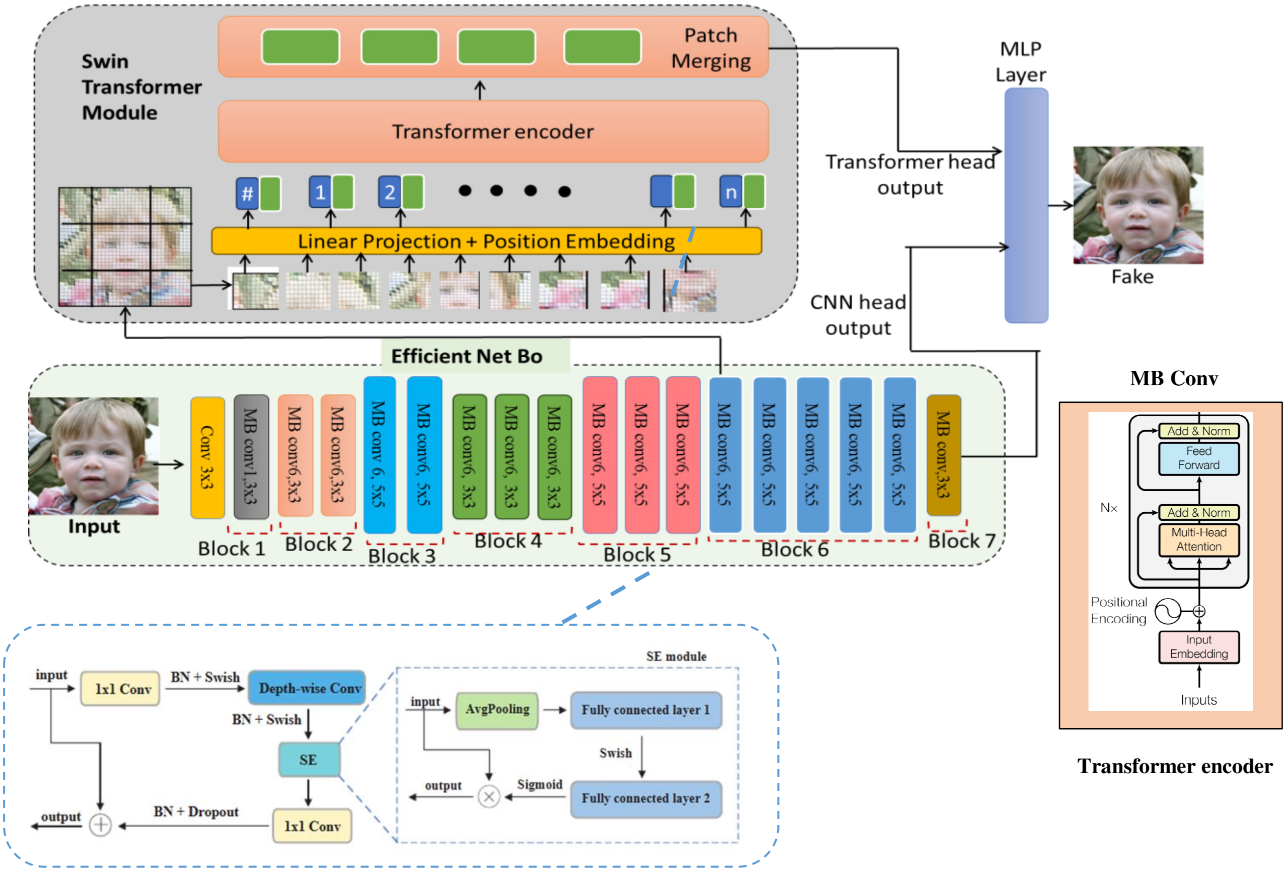

Fig. 1. Diagrammatic representation of the proposed EffiSwinNet

et al [23] focus on three convolutional neural network (CNN) algorithms—VGG16, VGG19, and ResNet—for this purpose. To evaluate the performance of these CNN models, a dataset comprising 1,200 images, including a combination of fake and real images, was utilized. The deepfake images were generated using FaceApp, a prominent tool for creating manipulated visuals. Their findings demonstrate that VGG19 outperforms both VGG16 and ResNet50, achieving an impressive accuracy rate of 98% on the test dataset when small size is small. Another research by [24] uses GAN to generate deepfakes for dataset training. Numerous factors, such as facial expressions and image anomalies, are used to determine how dissimilar the deepfakes are from the actual ones. They uses EfficientNet B4 algorithm to identify a person's altered face in a frame. In terms of accuracy, the upgraded EfficientNet B4 performs better than baseline EfficientNet. Each frame's deepfake probability is computed, and the video's overall authenticity is determined. With a detection rate of over 92%. With this motivation we proposed a cross architecture combining vision transformers with baseline CNN based model for superior detection accuracy, reduced computational complexity, and strong generalization.

III. METHODOLOGY

TABLE I. DISTRIBUTION OF DATASET

Dataset/ Class	Real	Fake
Test	2500	2500
Train	12500	12500
Valid	2500	2500

A. Data and Pre-processing

To effectively develop and evaluate a DeepFake detection algorithm, two key criteria must be met: the model must be trained on a large-scale dataset, and the dataset should closely reflect the quality and diversity of DeepFake content found online. In this study, we utilize a dataset comprising 140,000 real and fake facial images, 70,000 authentic images, sourced from the benchmarked DFDC (Deepfake detection challenge) data released by Meta [25], and 70,000 synthetic images generated using StyleGAN. The real images are originally collected from Flickr, which also employs StyleGAN-based filtering. The dataset encompasses a wide range of demographics, including individuals of various ethnicities, age groups, and the presence of facial accessories such as sunglasses and hats. Basic preprocessing has already been performed, including face detection and cropping using the dlib library. To further enhance the dataset's diversity and improve model generalization, data augmentation techniques such as random horizontal flipping and normalization are applied. While the original images are sized at 256×256 pixels, they are resized to 384 pixels to align with the input requirements of Swin Transformer . By Table 1 , originally the images were of size 256x256, which has been resized into 384 when fed into the Swin-Transformers. There were 12500 real and 12500 fake images for training , while 2500 real and fake images for testing purposes. This carefully curated and preprocessed dataset provides a balanced and diverse foundation for training and evaluating our proposed model, ensuring that it is both robust and capable of generalizing well to real-world deepfake detection scenarios.

B. Proposed Network EffiSwinNet

In this work, we propose EffiSwinNet, a hybrid deep learning model that integrates EfficientNetB0 with the Swin Transformer to enhance the performance of deepfake image detection. The overall block diagrams architecture, is illustrated in Figure 1. It is designed to combine the strengths of convolutional neural networks (CNNs) for local feature extraction and vision transformers for global contextual understanding. The model takes as input a preprocessed RGB image, resized to match the expected dimensions of the chosen Swin Transformer variant 384×384. Preprocessing steps include normalization and random horizontal flipping to improve generalization. The input image is first passed through the EfficientNetB0 backbone, a lightweight CNN architecture known for its efficiency and accuracy. EfficientNetB0 is composed of a series of MBConv blocks with squeeze-and-excitation (SE) modules and Swish activation functions as in [26]. To capture spatially meaningful features without losing semantic richness, we extract an intermediate feature map from the first MBConv block of stage 6. This layer is chosen as it provides a good balance between spatial resolution and high-level semantic information, making it suitable for transformer-based processing.

In parallel, the final output of EfficientNetB0 is fed into a CNN classification head, consisting of a global average pooling layer followed by dropout regularization. This head is responsible for capturing high-level features that are directly useful for classification tasks. Simultaneously, the intermediate feature map from the earlier EfficientNet layer is passed through the Swin Transformer branch, which includes patch extraction, embedding, and shifted window-based self-attention. The Swin Transformer operates on non-overlapping patches to efficiently model long-range dependencies. Positional embeddings are added to maintain spatial coherence, and the output features are hierarchically reduced through patch merging to obtain a compact yet informative representation. This output is then processed by a Transformer head composed of global average pooling, dropout, and batch normalization layers.

The outputs from both the CNN and Swin Transformer heads are then concatenated to form a unified feature vector, which is passed through a final Multilayer Perceptron (MLP) classification layer consist of 2 dense layers for classification with Gaussian Error Linear Unit (GELU) activation function. This layer outputs the prediction logits indicating whether the input image is real or fake. Additionally, during inference, the model can return intermediate outputs from both the CNN and Transformer branches. These outputs can be used in visual interpretation methods such as Grad-CAM to analyze the attention and decision patterns of the model, thereby improving explainability.

C. Training of EffiSwinNet and other variants

When it comes to deepfake detection, CNN models such as EfficientNet have demonstrated state-of-the-art performance. While CNNs are highly effective at extracting hierarchical local features, the Swin Transformer shows strong potential for interpreting complex visual data by leveraging self-attention mechanisms and visual attention maps. In this study, we introduce a hybrid architecture—EffiSwinNet—which combines the strengths of both models. Specifically, feature maps are extracted from the EfficientNet-B0 model up to the first MBConv layer of stage 6, and these intermediate features are then passed to the Swin Transformer module for global context modeling. The hyperparameters of EffiSwinNet are detailed in Table II. In alignment with EfficientNet-B0 standards, input images are resized to 384×384 pixels. After convolution, the intermediate feature map output has a spatial resolution of 24, which serves as input to the Swin Transformer. The model utilizes ImageNet-1K pre-trained weights for EfficientNet-B0 and incorporates 128 MLP layers in the Swin Transformer. It is configured with a window size of 2, shift size of 1, patch size of 2×2, eight attention heads, and a learning rate of 0.5. Although a learning rate of 0.5 may appear unusually high in the context of deep learning, as, both the EfficientNetB0 and Swin Transformer components were initialized using ImageNet-1K pretrained weights. This allowed us to perform aggressive fine-tuning, rather than training from scratch, enabling faster convergence with higher learning rates. Additionally, the use of a relatively small batch size of 8 supports the employment of larger learning rates, as smaller batches often introduce more noise during gradient updates, which can be balanced by higher learning rates to facilitate learning stability and speed. The training process is conducted over 20 epochs with a batch size of 8, resulting in a total of 4,258,246 trainable parameters.

For comparative analysis, we also implemented several variants of the Swin Transformer: Swin-Base, Swin-Tiny, and Swin-Small. The Swin-Base variant uses an input size of 128×128×3, with a window size of 8, a patch size of 4, a batch size of 32, and a learning rate of 2e-5. It was trained on 25,000 images over 200 epochs and tested on 5,000 images, with a total of 4,225,566 trainable parameters. Additionally, we employed pre-trained Swin-Tiny and Swin – Small models from Microsoft , both fine tuned on the deepfake dataset using ImageNet-1K weights. Although both models share the same learning rate of 5e-4, Swin-Tiny uses a batch size of 128, while Swin-Small uses a batch size of 64. Due to its deeper and wider architecture, Swin-Small has nearly twice as many trainable parameters (4.87M) compared to Swin-Tiny (2.75M). Consequently, Swin-Tiny required 100 training epochs, whereas Swin-Small required 150 epochs. These configurations and model variants facilitate a comprehensive evaluation of our proposed EffiSwinNet, demonstrating its effectiveness and computational efficiency in comparison to standard Swin Transformer architectures for deepfake image detection.

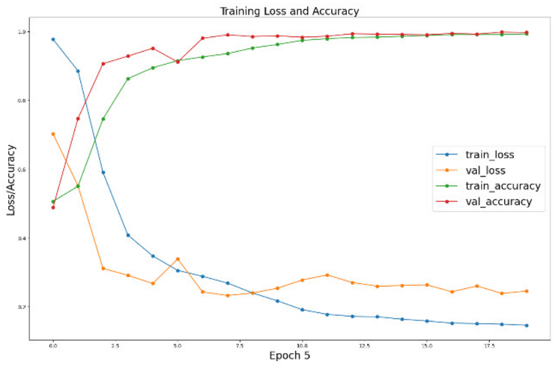

Fig.2. Accuracy and Loss curves for EffiSwinNet

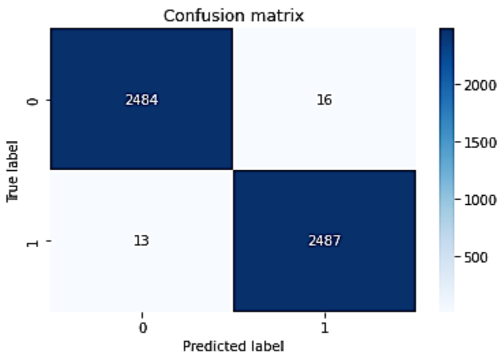

Fig.3. Confusion matrix plot for true and predicted labels with EffiSwinNet

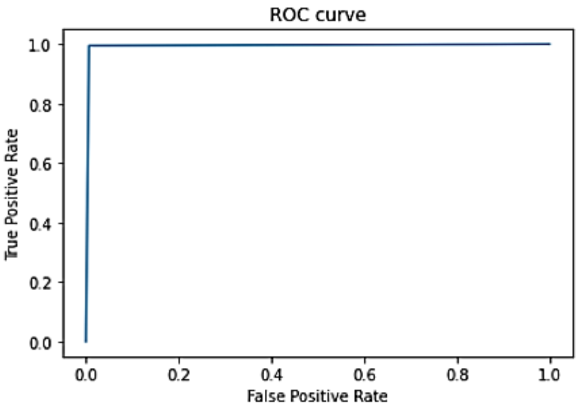

Fig.4. AUC-ROC curve for EffiSwinNet

TABLE II. HYPER-PARAMETERS FOR EFFISWINNET

Parameters	Values
Number of MLP Layers	128
Patch Size	2x2
Shifting Window Size	1
Attention Window Size	2
Number of Attention Heads	8
Learning Rate	0.5
Trainable Parameters	4,258,246

IV. RESULTS AND DISCUSSION

Among all the experimental models, the proposed Hybrid Swin Transformer model, which integrates the EfficientNet-B0 and Swin Transformer architectures, achieved the most superior performance. As depicted in Figure 2, the model attained a classification accuracy of 99% and a cross-entropy loss of 0.14, demonstrating a significant advancement over baseline CNN and transformer-only .EffiSwinNet was evaluated on a balanced test set comprising 5,000 facial images (2,500 real and 2,500 deepfakes).

The confusion matrix shown in Figure 3 presents the classification results: the model correctly identified 2,487 fake images (true positives) and 2,484 real images (true negatives). Only 13 fake images were misclassified as real (false negatives), and 16 real images were labelled as fake (false positives). The low error rate further supports the model's robustness and reliability in real-world deepfake detection tasks.

In addition, the model's discriminative power is validated by the Receiver Operating Characteristic (ROC) curve, illustrated in Figure 4, which shows an Area Under the Curve (AUC) of 0.9942. This near-perfect AUC score signifies excellent sensitivity and specificity, with a very high true positive rate and minimal false positive rate. The steep curve rising toward the top-left corner indicates the model’s strong capacity to differentiate between real and fake samples across varying decision thresholds.

A comparative analysis was conducted with other Swin Transformer variants, including the Base Swin Transformer, Swin-Tiny, and Swin-Small (Table III). While all models achieved commendable results, the proposed Hybrid Swin Transformer achieved the highest values across all metrics—Precision (99%), Recall (99%), F1-Score (99.42%), and Accuracy (99%)—despite being trained for only 20 epochs, in contrast to the 100–200 epochs required by other models. This result highlights the hybrid model's training efficiency and parameter optimization.

Finally, To further analyse the model's interpretability, the Grad-CAM (Gradient-weighted Class Activation Mapping) technique was employed to generate visual attention heatmaps (Figure 7). These heatmaps reveal that the EffiSwinNet model effectively focuses on both local discriminative regions and global spatial dependencies within the input image—something traditional CNNs struggle with. The attention heatmaps generated by the EffiswinNet model (Figure 7) illustrate its ability to attend to both fine-grained and global facial regions. Unlike traditional CNNs, which focus narrowly on local texture patterns, the EffiswinNet effectively combines self-attention and convolutional features, enabling the model to localize and interpret subtle deepfake artifacts distributed across the entire image. Moreover, the proposed EffiSwinNet model comprises approximately 4.25 million trainable parameters, ensuring a balanced trade-off between performance and resource usage. The average inference time per image is measured to be around 43 milliseconds on an NVIDIA RTX 3060 GPU, making it suitable for near real-time applications. Additionally, the estimated computational load is about 2.8 GFLOPs, and the memory usage during inference is approximately 600 MB per image. These results indicate that the model is lightweight and efficient enough to be deployed on moderately powered GPU systems without significant resource constraints.

V. CONCLUSION

Swin-Transformers were used to detect deepfake photos using three alternative methods, with a balanced dataset of 25,000 images used for training. The first method achieved an accuracy of 88.57% by training a base Swin-Transformer model from scratch. The Swin-Tiny and Swin-Small models were improved using more MLP layers in the second method, which included transfer learning. Swin-Tiny performed better than Swin-Small among them, with an accuracy of 91.13%. The third and last method produced a hybrid model that merged a CNN with a Swin-Transformer, achieving an astounding 99% accuracy rate. Experiments on 5,000 photos showed that this hybrid model performed exceptionally well, with 99% precision and 99% recall. Grad-CAM-based heatmaps demonstrated the model's capacity to use both local and global attention to highlight important image characteristics. The hybrid method produced quicker training times even though it needed training settings that were comparable to those of the Swin-Small model. The Swin-Transformer's worldwide refinement of the EfficientNet mid-layer feature maps is responsible for the performance boost.

Fig. 2. Heat map showing the areas of attention.

TABLE III. HYPER-PARAMETERS FOR EFFISWINNET

Model	Epoch	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
Base Swin	200	88	87	88	88.5
Swin-Tiny	100	88	94	91.	91.13
Swin-Small	150	87	91	89.13	89.14
EffiSwinNet	**20**	**99**	**99**	**99.42**	**99**

Even while the outcomes were quite encouraging, they may still be better. The majority of the deepfake images in the current dataset were produced with the StyleGAN approach. A more varied dataset (like Celeb-DF, FaceForensics++) can be included to validate the generalization. Furthermore, even though deepfake videos are common, the study only looked at deepfake photos. Video-based datasets were not investigated due to time constraints. Extending these models to identify deepfakes in videos may be the main goal of future research. Last but not least, cross-validation with a completely different dataset was not carried out because the testing dataset employed in this work was taken from the same dataset as the training set, providing room for more research.

REFERENCES

[1] T. Zhang, "Deepfake generation and detection, a survey," *Multimedia Tools and Applications*, vol. 81, no. 5, pp. 6259–6276, Jan. 2022, doi: https://doi.org/10.1007/s11042-021-11733-y

[2] A. Ghai, P. Kumar, and S. Gupta, "A deep-learning-based image forgery detection framework for controlling the spread of misinformation," *Information Technology & People*, vol. ahead-of-print, no. ahead-of-print, Jun. 2021, doi: https://doi.org/10.1108/itp-10-2020-0699.

[3] H. Samer, Al-Khazraji, H. Saleh, A. Khalıd, and I. Mıshkhal, "Impact of Deepfake Technology on Social Media: Detection, Misinformation and Societal Implications," Technology and Science, vol. 23, 2023, Available: http://www.epstem.net/tr/download/article-file/3456697.

[4] C. Okolie, "Artificial Intelligence-Altered Videos (Deepfakes), Image-Based Sexual Abuse, and Data Privacy Concerns," *Journal of International Women's Studies*, vol. 25, no. 2, Mar. 2023, Available: https://vc.bridgew.edu/jiws/vol25/iss2/11/

[5] R. Khan, Mohd Taqi, and A. Afzal, "Deepfakes in Finance," Advances in information security, privacy, and ethics book series, pp. 91–120, Jul. 2024, doi: https://doi.org/10.4018/979-8-3693-5298-4.ch006.

[6] B. Adithya, J. Ranjith, J. Poojitha, B. S. Kumar, and Griddaluri Aashish, "A Survey on Detection of Deepfake Text and Sentiment Analysis using Machine Learning Models," CRC Press eBooks, pp. 83–107, Feb. 2025, doi: https://doi.org/10.1201/9781003504832-8.

[7] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., ... & Bengio, Y. (2020). I. Goodfellow et al., "Generative adversarial networks," Communications of the ACM, vol. 63, no. 11, pp. 139–144, Oct. 2020, doi: https://doi.org/10.1145/3422622

[8] Preeti, M. Kumar, and H. K. Sharma, "A GAN-Based Model of Deepfake Detection in Social Media," Procedia Computer Science, vol. 218, pp. 2153–2162, Jan. 2023, doi: https://doi.org/10.1016/j.procs.2023.01.191.

[9] A. Melnik et al., "Face Generation and Editing with StyleGAN: A Survey," IEEE Transactions on Pattern Analysis and Machine Intelligence, pp. 1–21, Jan. 2024, doi: https://doi.org/10.1109/tpami.2024.3350004.

[10] S. Takahashi et al., "Comparison of Vision Transformers and Convolutional Neural Networks in Medical Image Analysis: A Systematic Review," Journal of Medical Systems, vol. 48, no. 1, Sep. 2024, doi: https://doi.org/10.1007/s10916-024-02105-8.

[11] K. U. Rehman et al., "A Feature Fusion Attention-based Deep Learning Algorithm for Mammographic Architectural Distortion Classification," IEEE Journal of Biomedical and Health Informatics, pp. 1–12, 2025, doi: https://doi.org/10.1109/jbhi.2025.3547263.

[12] H. Yunusa, S. Qin, C. Abdulrahman, A. A. Yusuf, I. Bello, and A. Lawan, "Exploring the Synergies of Hybrid CNNs and ViTs Architectures for Computer Vision: A survey," *arXiv.org*, 2024. https://arxiv.org/abs/2402.02941.

[13] L. Gai, M. Xing, W. Chen, Y. Zhang, and X. Qiao, "Comparing CNN-based and transformer-based models for identifying lung cancer: which is more effective?," Multimedia Tools and Applications, vol. 83, no. 20, pp. 59253–59269, Dec. 2023, doi: https://doi.org/10.1007/s11042-023-17644-4.

[14] Ishak Pacal, Melek Alaftekin, and Ferhat Devrim Zengul, "Enhancing Skin Cancer Diagnosis Using Swin Transformer with Hybrid Shifted Window-Based Multi-head Self-attention and SwiGLU-Based MLP," Journal of Imaging Informatics in Medicine, Jun. 2024, doi: https://doi.org/10.1007/s10278-024-01140-8.

[15] G. A. Pereira and M. Hussain, "A Review of Transformer-Based Models for Computer Vision Tasks: Capturing Global Context and Spatial Relationships," arXiv.org, 2024. https://arxiv.org/abs/2408.15178.

[16] A. J. Xi and E. Chen, "Classifying Deepfakes Using Swin Transformers," arXiv.org, 2025. https://arxiv.org/abs/2501.15656.

[17] Talluri Harshitha, T. Simhadri, Thadakuluru Jaswanthi, N. Rayvanth, and R. P. Singh, "Deepfake Image Detection Using Light-Weight Attention Integrated MobileNetV3 Model," Lecture notes in computer science, pp. 130–142, Jan. 2025, doi: https://doi.org/10.1007/978-3-031-84543-7_11

[18] R. P. Singh, N. H. Sree, K. L. S. P. Reddy, and K. Jashwanth, "Convergence of Deep Learning and Forensic Methodologies Using Self-attention Integrated EfficientNet Model for Deep Fake Detection," SN Computer Science, vol. 5, no. 8, Dec. 2024, doi: https://doi.org/10.1007/s42979-024-03455-3.

[19] M. Nawaz, A. Javed, and A. Irtaza, "A deep learning model for FaceSwap and face-reenactment deepfakes detection," Applied Soft Computing, vol. 162, p. 111854, Sep. 2024, doi: https://doi.org/10.1016/j.asoc.2024.111854.

[20] M. Shahin and M. Deriche, "A Novel Framework based on a Hybrid Vision Transformer and Deep Neural Network for Deepfake Detection," pp. 329–333, Apr. 2024, doi: https://doi.org/10.1109/ssd61670.2024.10548578.

[21] S. Usmani, S. Kumar, and Debanjan Sadhya, "Efficient deepfake detection using shallow vision transformer," Multimedia Tools and Applications, vol. 83, no. 4, pp. 1–24, Jun. 2023, doi: https://doi.org/10.1007/s11042-023-15910-z

[22] S. R. Mishra, H. Mohapatra, and Mahendra Kumar Gourisaria, "A Robust Approach for Deepfake Detection Using SWIN Transformer," Research Square (Research Square), Jul. 2024, doi: https://doi.org/10.21203/rs.3.rs-4672886/v1.

[23] Zahra Nazemi Ashani, "Comparative Analysis of Deepfake Image Detection Method Using VGG16, VGG19 and ResNet50," Journal of Advanced Research in Applied Sciences and Engineering Technology, vol. 47, no. 1, pp. 16–28, Jun. 2024, doi: https://doi.org/10.37934/araset.47.1.1628.

[24] V. Gowri Priyaa, M. Jaya Harrish, M. Udhayakumar, N. Jothieswaran, and K. Dinesh, "Efficientnet-Based Deep Learning Approach for Video Forgery Detection and Authentication," 2024 10th International Conference on Communication and Signal Processing (ICCSP), pp. 334–338, Apr. 2024, doi: https://doi.org/10.1109/iccsp60870.2024.10544113.

[25] B. Dolhansky et al., "The DeepFake Detection Challenge Dataset," arXiv:2006.07397 [cs], Jun. 2020, Available: https://arxiv.org/abs/2006.07397

[26] W. Zhou, J. Ji, Y. Jiang, J. Wang, Q. Qi, and Y. Yi, "EARDS: EfficientNet and attention-based residual depth-wise separable convolution for joint OD and OC segmentation," Frontiers in Neuroscience, vol. 17, Mar. 2023, doi: https://doi.org/10.3389/fnins.2023.1139181